

Soru 1

Which is a valid property of the standard deviation?

- a. It can never be zero
- b. It can only be calculated for normal distributions
- c. It measures the clustering of data around the mean
- d. It is smaller than the variance
- e. It equals the interquartile range

Soru 2

If events A and B are *mutually exclusive*, then:

- a. $P(A \cap B) = 0$
- b. $P(A \cup B) = P(A) P(B)$
- c. $P(A \cap B) = P(A) \times P(B)$
- d. $P(A | B) = P(A)$
- e. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Soru 3

Which of the following statements is true for *independent* events A and B?

- a. $P(A \cap B) = 0$
- b. $P(A \cup B) = P(A) + P(B)$
- c. $P(A | B) = P(A)$
- d. $P(A) + P(B) = 1$
- e. They cannot occur at the same time.

Soru 4

If $P(A) = 0.4$ and $P(B) = 0.5$, and events A and B are independent, what is $P(A \cap B)$?

- a. 0.0
- b. 0.1
- c. 0.4
- d. 0.45
- e. 0.20

Soru 5

If the sample space consists of equally likely outcomes, the probability of an event is:

- a. The total number of outcomes
- b. The ratio of favorable outcomes to total outcomes
- c. Always 1
- d. The ratio of total outcomes to favorable outcomes
- e. Always 0

Soru 6

Which of the following is an example of a *conditional probability*?

- a. $P(A)$
- b. $P(A \cap B)$
- c. $P(A \cup B)$
- d. $P(A | B)$
- e. $P(A')$

Soru 7

Which of the following statements about a discrete random variable is true?

- a. It can take on infinitely many values within an interval
- b. It is always uniformly distributed
- c. It takes on countable possible values
- d. It has no associated probability distribution
- e. It is the same as a continuous random variable

Soru 8

The mean of a binomial random variable X with parameters n and p is:

- a. $n \times p$
- b. $n \times (1 - p)$
- c. $\sqrt{n p (1-p)}$
- d. $(np) / (n + p)$
- e. n^2

Soru 9

Which distribution is often used to model the number of occurrences in a fixed interval of time or space?

- a. Binomial distribution
- b. Poisson distribution
- c. Geometric distribution
- d. Hypergeometric distribution
- e. Uniform distribution

Soru 10

A sampling distribution describes the distribution of which of the following?

- a. Individual data points from the population
- b. Observed data from non-random samples
- c. A statistic (e.g., sample mean) across many samples
- d. Parameter values for a single sample
- e. Experimental errors in measurement

Soru 11

A 95% confidence interval means:

- a. The true parameter will definitely lie in the calculated interval
- b. 95% of the sample data lie within the interval
- c. In 95% of repeated samples, the interval will contain the true parameter
- d. The confidence interval is valid only for normal distributions
- e. The point estimate is 95% accurate

Soru 12

A student uses a small sample ($n=10$) and does *not* know the population standard deviation. The appropriate distribution to use for constructing a confidence interval for the mean is:

- a. t-distribution
- b. z-distribution
- c. F-distribution

- d. Chi-square distribution
- e. Binomial distribution

Soru 13

Which of the following is *not* typically considered part of the statistical thinking process?

- a. Identify the population and relevant variables
- b. Compare theoretical distributions to observed data
- c. Summarize data using numerical and graphical methods
- d. Randomly guess conclusions without looking at data
- e. Use data to make inferences about a population

Soru 14

When conducting a hypothesis test for a population mean with an unknown population standard deviation (and assuming normal population), the test statistic is typically:

- a. $z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$
- b. $\chi^2 = ((n-1) s^2) / \sigma^2$
- c. $t = (\bar{X} - \mu_0) / (s / \sqrt{n})$
- d. $F = MSR / MSE$
- e. $z = (\bar{X} - \mu_0) / (s / \sqrt{n})$

Soru 15

A population of test scores is normally distributed with mean $\mu = 80$ and standard deviation $\sigma = 10$. What is the probability that a single randomly chosen score from this population is greater than 92?

- a. 0.0515
- b. 0.1151
- c. 0.5000
- d. 0.8849
- e. 0.9465

Soru 16

Let X be a binomial random variable with $n=8$ and $p=0.20$. What is the probability that $X = 2$?

- a. 0.044
- b. 0.134
- c. 0.167
- d. 0.2936
- e. 0.450

Soru 17

A random sample of size $n=25$ from a population with known $\sigma = 4$ yields $\bar{X} = 50$. Construct a 95% confidence interval for μ .

- a. (48.43, 51.57)
- b. (47.63, 52.37)
- c. (49.25, 50.75)
- d. (50.00, 52.00)
- e. (45.00, 55.00)

Soru 18

A random sample of size $n=10$ from a normal population gives $\bar{X} = 22.5$ and $s = 2.3$. Construct a 99% confidence interval for μ . ($t_{0.005,9} \approx 3.25$)

- a. (19.7, 25.3)
- b. (20.1, 24.9)
- c. (18.3, 26.7)
- d. (20.5, 24.5)
- e. (21.3, 24.1)

Soru 19

A manufacturer claims $\mu \geq 40$ hours for battery life. $\sigma = 6$. From a sample of $n=36$, $\bar{X} = 38.5$. At $\alpha = 0.05$, test $H_0: \mu \geq 40$ vs. $H_a: \mu < 40$. The test statistic is:

- a. -1.00
- b. -1.25
- c. -1.50
- d. -1.75
- e. -2.50

Soru 20

A dietitian claims $\mu = 500$ mg sodium. From $n=9$ servings, $\bar{X} = 520$, $s=30$. Assume normality; at $\alpha=0.05$, test $H_0: \mu=500$ vs. $H_a: \mu \neq 500$. The test statistic is:

- a. 1.65
- b. 2.00
- c. 2.31
- d. 2.36
- e. 3.12

Soru 21

An economist wants to estimate mean monthly grocery expenditure to within $\pm \$10$ at 95% confidence. A pilot study suggests $\sigma \approx \$40$. What sample size is needed?

- a. 25
- b. 49
- c. 62
- d. 78
- e. 100

Soru 22

A population has $\mu=70$, $\sigma=12$. A sample of size $n=36$ is drawn. What is $P(\bar{X} > 75)$?

- a. 0.9938
- b. 0.0062
- c. 0.0228
- d. 0.0787
- e. 0.1100

Soru 23

A random sample of 200 voters finds 120 who support a candidate. Construct a 95% confidence interval for the true proportion p .

- a. (0.52, 0.68)

- b. (0.45, 0.75)
- c. (0.53, 0.67)
- d. (0.58, 0.68)
- e. (0.48, 0.72)

Soru 24

A distribution is called *right-skewed* if it has:

- a. A long tail extending to the left
- b. A long tail extending to the right
- c. Two distinct peaks
- d. A symmetrical shape
- e. No peak at all

Soru 25

If a dataset has a mean of 50 and a median of 45, which statement is most likely correct about the data's shape?

- a. The distribution is symmetric
- b. The distribution is right-skewed
- c. The distribution is left-skewed
- d. The distribution is bimodal
- e. The shape cannot be inferred from this information